SE4031 – Games Development

INTERACTIVE MEDIA
IMAGINE. CREATE. ENTERTAIN.

Assignment 1
2D Game Document

Game Title – DECK 04

By: Amarasekara P E – IT22347008

Table of Contents

Overview 3
Implementation of Core Features 3
Screenshots 5
Control Guide 7
Creative Feature 8
Credits 8
Video Link 8

Overview

Deck 04 is a 2d side scrolling sci fi shooter inspired by Jetpack Joyride and Mass Effect. It involves the player having to navigate the titular Deck 4 aboard their spaceship dodging obstacles and destroying enemy robots. Players can fire their guns which send out lasers used to destroy enemies. This is an endless runner where the game becomes faster as the players progress through the level. The core objective is to go as far as you can without dying and get a high score. Players have a health bar which decreases if an obstacle or an enemy hits them. However, spawning throughout the level are health packs which can be picked up by the player to restore a fraction of their health. Players can also pause the game as they please where they can either resume the game or go back to main menu. At the game over screen players display their final score and can either restart the game or return to the main menu. In the main menu players can view the instructions regarding how to play the game.

Implementation of Core Features

1. The Jetpack (Physics & Movement)

Instead of just teleporting the player up and down, the game treats the player like a physical object with weight.

- **The Motor:** When you hold Space, a "motor" pushes the player upward. We don't just move them; we give them upward speed (velocity).
- **The Weight:** When you let go, the game's built-in gravity takes over and pulls you back down to the floor naturally.
- **The Look:** The game constantly checks: "Is the player touching the floor?" If no, play the flying animation. If yes, play the walking animation.

2. The "Flash" Effect (Timing & Visuals)

When you get hit, you want the player to turn red for a moment. But if the game "waited" for that second to pass, the whole screen would freeze.

- **The Stopwatch:** We use a special background timer. When you hit an obstacle, the player turns red and a "mini-stopwatch" starts.
- **Multitasking:** While that stopwatch is counting to 1 second, the rest of the game keeps running—enemies keep moving, and you can still fly. Once the stopwatch hits zero, it flips the player's color back to normal.

3. The "Brain" (The Game Manager)

The GameManager is like the conductor of an orchestra. It doesn't play the instruments, but it tells everyone else what to do.

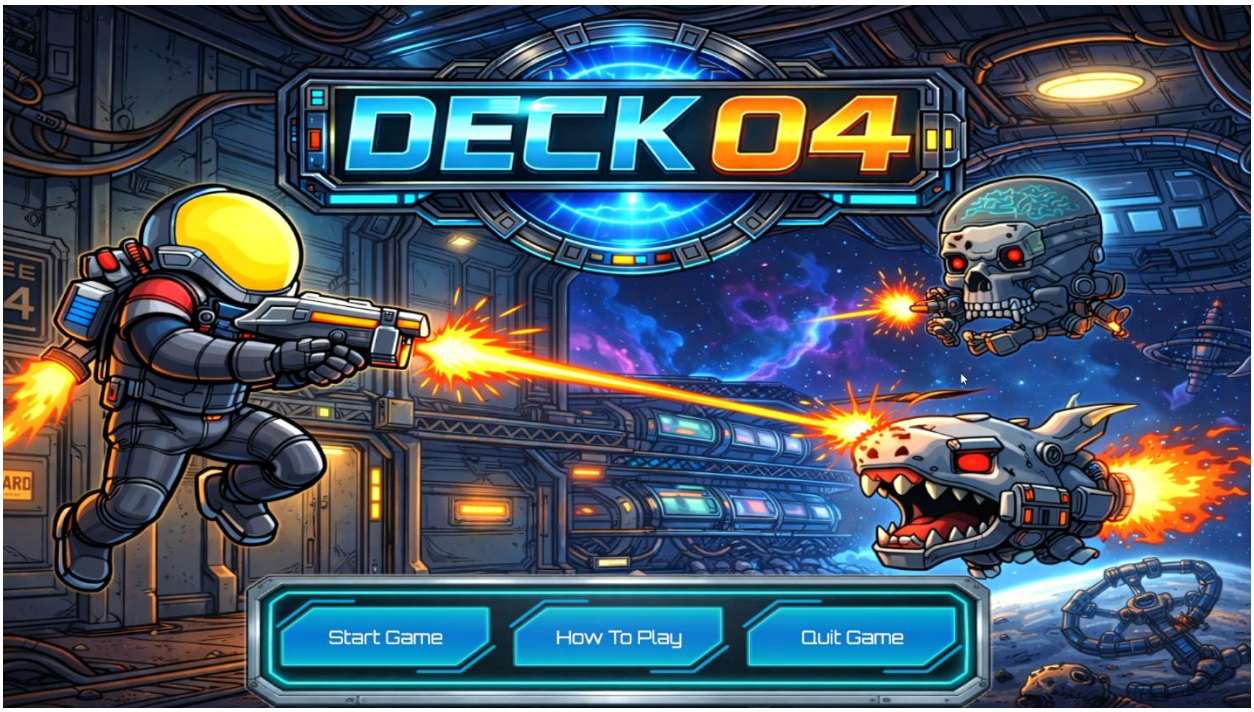
- **Speeding Up:** The "Brain" keeps a master speed dial. Every second you survive, it turns the dial up a little bit. Every enemy and obstacle looks at this dial to see how fast they should be moving toward you.
- **The Death Sequence:** When you die, the Brain does three things in a row:
    1. **Freeze:** It stops time for everything else (enemies stop moving).
    2. **Boom:** It lets the explosion animation play (this is the only thing allowed to move while time is frozen).
    3. **The Menu:** It waits for the explosion to finish before showing you the "Game Over" screen.

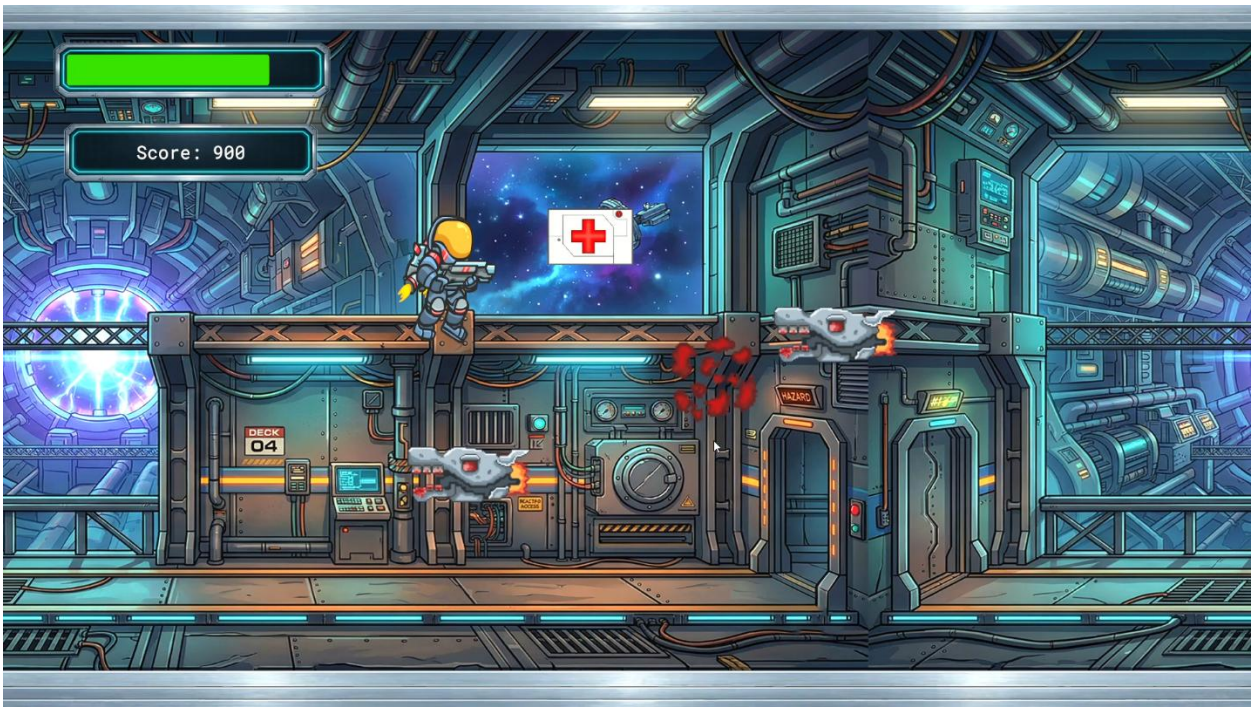
4. The Sound System (One-offs vs. Loops)

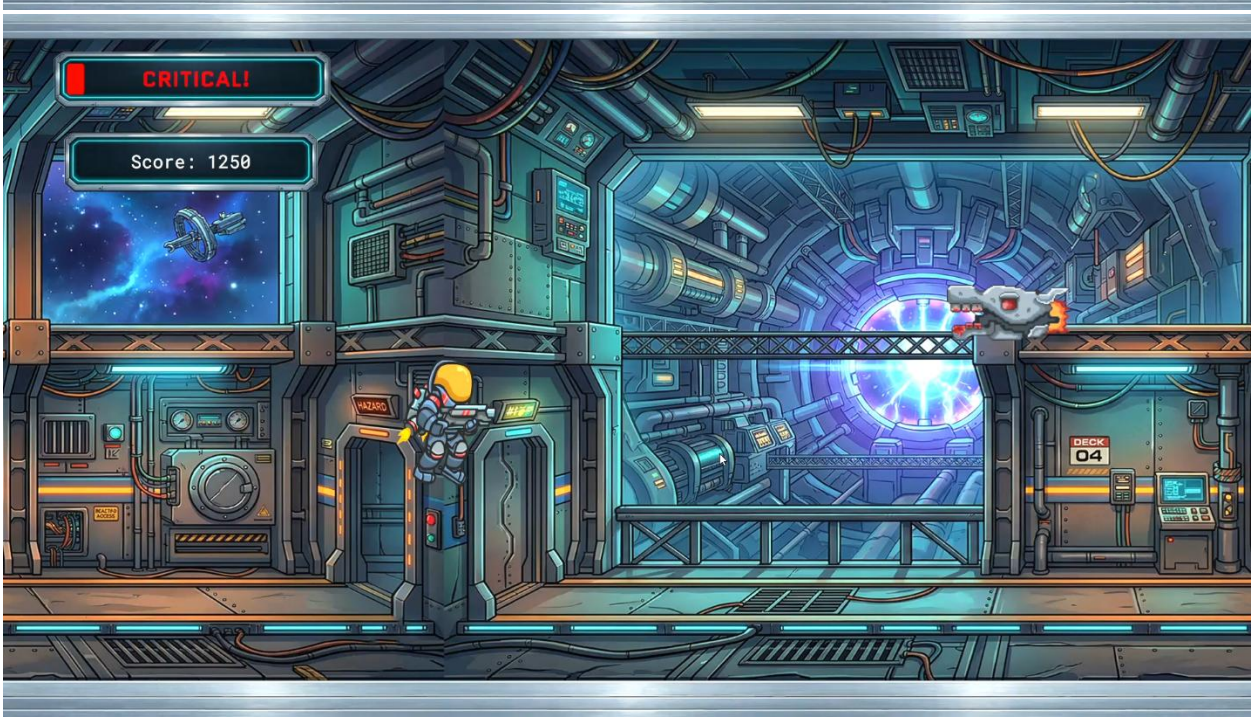
We handle sounds differently depending on how they are used so the audio doesn't sound "glitchy."

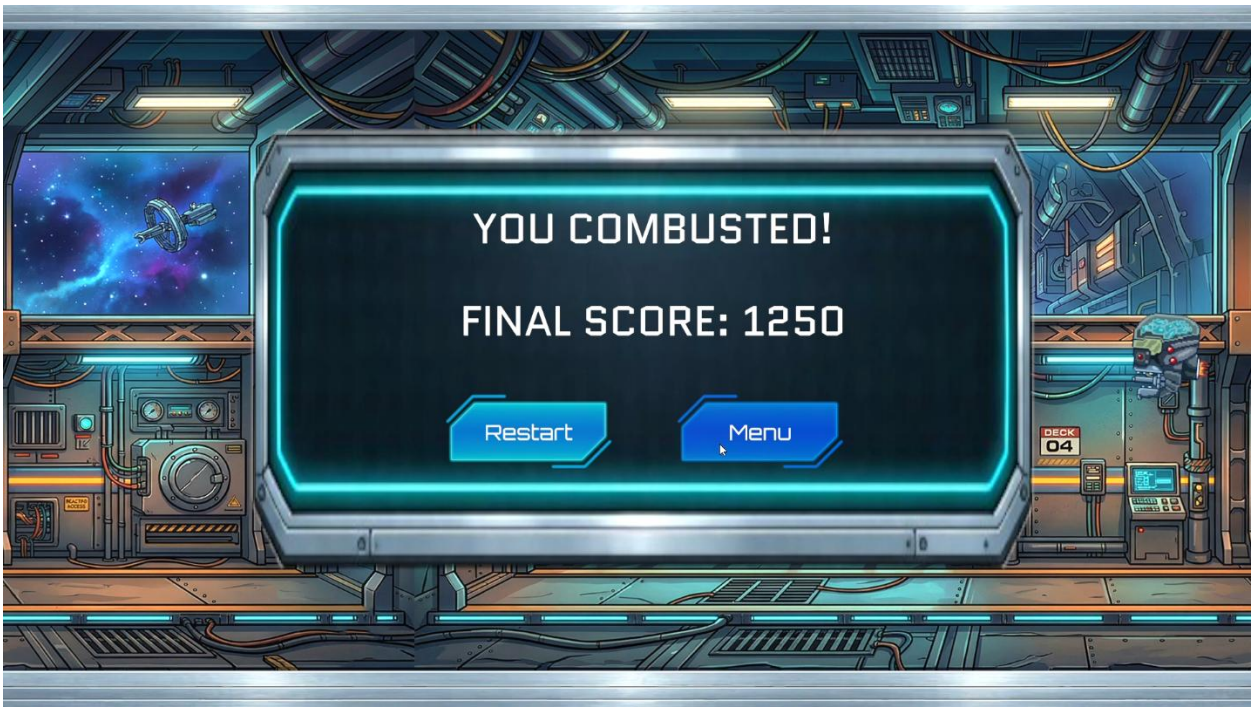
- **One-Offs:** Sounds like a laser blast or an explosion are "fire and forget." You trigger them, they play once, and they disappear.
- **The Loop:** Sounds like the jetpack engine or footsteps are "toggles." They stay "ON" as long as you are holding the button and turn "OFF" the moment you let go.
- **The Pause:** We made sure that if you open the menu, the sounds "Pause" instead of stopping. This way, when you resume, the jetpack sound doesn't have to start over from the beginning.

Screenshots




Score: 900
DECK
04
HAZARD

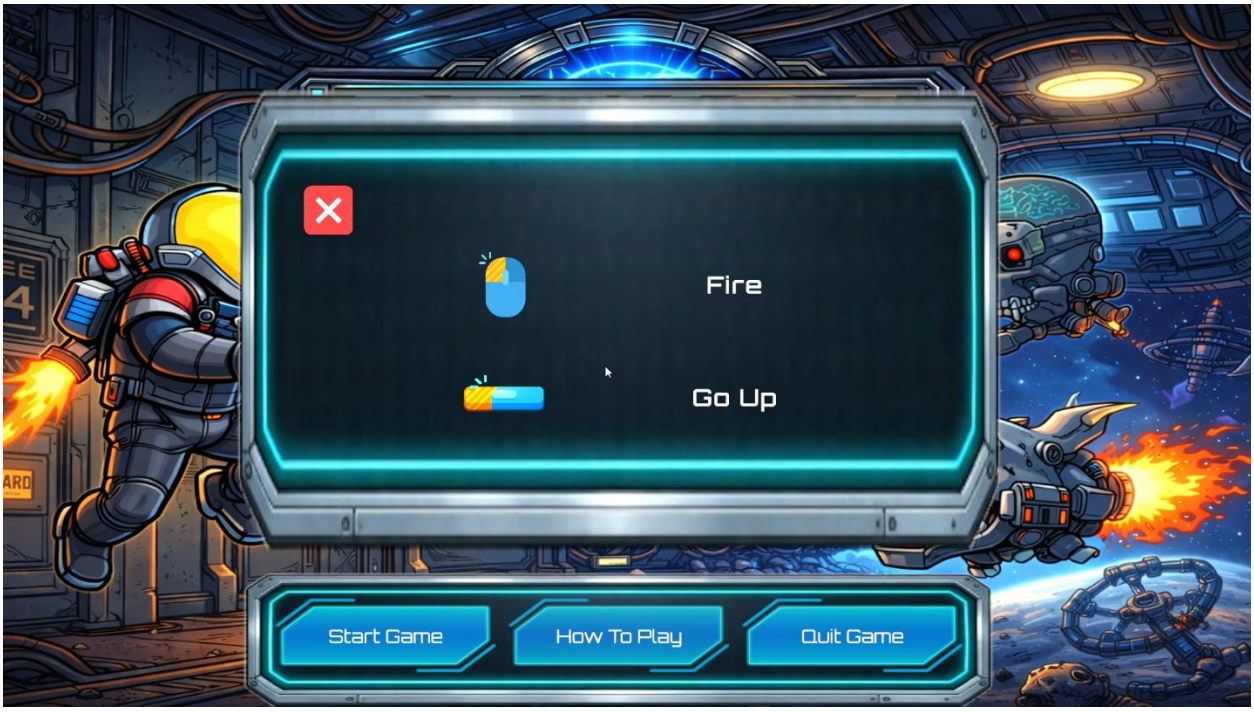

CRITICAL!
Score: 1250
HAZARD
DECK
04



Control Guide

Keyboard/Spacebar – Thrust upward

Mouse/Left Click – Fire Gun



Creative Feature

AI Behavior: The Tracer Unit

The Tracer Unit serves as the primary pressure mechanic in the game. While common enemies simply obstruct the player's path, the Tracer forces active engagement by constantly recalculating its flight path to intercept the player's current coordinates.

1. Target Acquisition (Dynamic Tracking)

The Tracer does not follow a pre-set path. Instead, it maintains a constant "mental link" with the player.

- **The Logic:** Every single frame, the unit compares its own location to the player's location. It calculates the most direct line between these two points, ensuring that as you move your jetpack up or down, the Tracer immediately adjusts its angle to follow.

2. Constant Pursuit Speed (Standardized Momentum)

A common issue with homing AI is that they tend to move faster when they are far away and slower when they are close. The Tracer is programmed to avoid this.

- **The Logic:** We use a "Speed Regulator" (Normalization). This ensures that whether the Tracer is across the screen or right next to your hull, it moves at a perfectly consistent, predictable speed. This allows the player to learn the Tracer's timing and dodge it reliably.

3. Omnidirectional Maneuvering

To ensure the Tracer feels like a high-tech drone, its movement is decoupled from its physical body.

- **The Logic:** The unit uses "Free-Space Movement." This means it can fly diagonally, vertically, or backward without needing to rotate its entire chassis first. This results in a "snappy," aggressive feel that makes the unit appear much more dangerous than standard patrol enemies.

Credits

Spaceman | 2D Characters | Unity Asset Store

Robot Shooting Game Sprite (Free) | 2D Environments | Unity Asset Store

Tank Constructor | 2D Textures & Materials | Unity Asset Store

2D Sci-Fi Weapons Pack | 2D Textures & Materials | Unity Asset Store

Sci-fi GUI skin | 2D GUI | Unity Asset Store

Video Link

https://drive.google.com/file/d/1Bp-x2zQzNk3AsIFLr8dEzgto64-RZ2Dp/view?usp=sharing